

# Functional Requirements Specification

## Change Request Management System

### 1. Objective

The system will provide a centralized platform for managing Clients, Projects, Change Requests, Approvals, Man-Hours, and Invoices.

Clients will be able to create and track Change Requests (CRs), while the company will manage estimation, implementation, delivery, and invoicing.

### 2. User Roles

#### Client

- Manage and view own CRs.
- View project information.
- Review estimates.
- Approve or reject CRs.
- Request clarification or rework.
- View invoices.

#### Company Admin

- Manage clients and projects.
- Manage CRs and estimates.
- Manage CR stages and man-hours.
- Manage approvals and delivery.
- Manage invoices.
- View assigned CRs.
- Update stage status.
- Record actual man-hours.
- Add comments.

### 3. Client & Project Management

The company can create and manage clients and their projects. Each project belongs to one client and can contain multiple CRs.

### 4. Change Request Management

Clients can create CRs under their projects.

Each CR includes:

- Name
- Description
- Priority
- Scope
- Attachments
- Comments

- Status

**Initial Status:** Pending Vendor Feedback

## 5. CR Status Management

The system will control status transitions. Only valid statuses for the current status can be selected.

| Current Status               | Available Statuses                                 |
|------------------------------|----------------------------------------------------|
| Pending Vendor Feedback      | Accepted / Rejected / Pending Client Clarification |
| Pending Client Clarification | Pending Vendor Feedback                            |
| Accepted                     | Estimation create                                  |
| Pending Customer Approval    | Estimate accepted / Rejected                       |
| Estimate accepted            | Analysis                                           |
| Analysis                     | Design                                             |
| Design                       | Development                                        |
| Development                  | Testing                                            |
| Testing                      | Pending Customer Approval                          |
| Rework Required              | Analysis / Design / Development / Testing          |
| Deployment                   | Delivered                                          |
| Delivered                    | Completed                                          |

## 6. CR Estimation

After a CR is accepted, the company creates an estimate containing:

- Total Estimated Man-Hours
- Man-Hour Rate
- Total Estimated Cost
- Expected Start Date
- Expected Finish Date

## 7. CR Execution

After customer approval, the CR moves through its implementation stages:

**Analysis → Design → Development → Testing**

## 8. Customer Approval & Rework

Customer approval is required before implementation and deployment.

If rework is requested:

- The client provides a reason.
- The relevant stage is selected.
- The CR returns to that stage.
- The CR continues through testing and approval again.

## 9. Delivery

After final customer approval:

**Deployment → Delivered → Completed**

The company can record basic deployment information.

## 10. Invoice Management

Completed CRs can be invoiced.

**Invoice relationship:** Client → Project → CR

Invoice statuses:

- Generated
- Sent
- Paid

Clients can view their invoices.

## 11. Client Portal

The client can view:

- Projects
- CRs
- CR Status
- CR Progress
- Estimated Hours
- Estimated Cost
- Approvals
- Invoices

## 12. Company Portal

The company can manage:

- Clients
- Projects
- CRs
- Estimates
- Stages
- Man-Hours
- Delivery
- Invoices
- Users

CRs can be filtered by:

- Client
- Project
- Status
- Priority
- Current Stage

## 13. Comments & Attachments

Users can add comments and attachments to CRs for:

- Requirements

- Clarifications
- Approvals
- Rework

## 14. Out of Scope

- Notifications
- Reporting & Statistics
- Audit / History Tracking
- Online Payment
- Accounting Integration
- Advanced Analytics

## 15. Main Workflow

**Create CR → Vendor Review → Estimate → Customer Approval → Implementation → Customer Approval → Rework if Required → Deployment → Delivered → Completed → Invoice**